MILITARY SPECIFICATION

TRANSISTOR, PNP, GERMANIUM
TYPE 2N987

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for a germanium, PNP, relatively high-gain transistor for use in high-frequency equipment circuits, and having the following significant characteristics at T_A = + 25°C. (See 3.2 herein.):

	P.G (at: f = 100 mc, V_{CB} = -6 Vdc, I_E = 1 mAdc)	P.G (at: f = 100 mc, V_{CB} = -4 Vdc, I_E = 1.5 mAdc)	h_{fe} (at: f = 1 kc, V_{CE} = -6 Vdc, I_C = 1 mAdc)	f_t (at: V_{CE} = -6 Vdc, I_E = 1 mAdc)	NF (at: f = 10.7 mc, V_{CB} = -6 Vdc, I_E = 1 mAdc; 1/)	NF (at: f = 100 mc, V_{CB} = -6 Vdc, I_E = 1 mAdc; 2/)
	db	db	---	mc	db	db
Minimum	10	10	60	40	---	---
Maximum	---	---	240	---	8	11

1/ Source impedance = 150 ohms 2/ Source impedance = 68 ohms

1.2 Absolute maximum ratings. -

P_C 1/	T_J	I_C	I_E	I_{ER}	V_{CB}	V_{CER}	V_{CEO}	V_{EB}	T_{stg}
mW	°C	mAdc	mAdc	mAdc	Vdc	Vdc	Vdc	Vdc	°C
65	100	-10	10	-1	-40	-40	-20	1	-55 to + 100

1/ This collector power dissipation is at T_A = + 25° C. For collector power dissipation at higher ambient temperatures up to + 100°C, derate in accordance with Figure 1 herein.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-19500 Semiconductor Devices, General Specification For

STANDARDS

MILITARY

MIL-STD-750 Test Methods For Semiconductor Devices

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Requirements. - Requirements for the transistor shall be in accordance with Specification MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. - The abbreviations and symbols used herein are defined in Specification MIL-S-19500 and as follows:

C_{re} . . . . . reverse capacity between collector and base of a transistor in a grounded emitter circuit

f_t . . . . . unity gain ($h_{fe} = 1$) frequency

P_C . . . . . collector power dissipation

P.G. . . . . available power gain (see formula in Fig. 3 herein.)

3.3 Design and construction. - The transistor shall be of the design, construction, and physical dimensions specified on Figure 2 herein.

3.3.1 Terminal-lead arrangement. - The terminal-lead arrangement on the transistor shall be as specified in Figure 2 herein.

3.3.2 Operating position. - The transistor shall be capable of proper operation in any position.

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3.4 Performance characteristics.- The transistor performance characteristics shall be as specified in Tables I and II herein.

3.5 Marking.- The transistor shall be marked in accordance with Specification MIL-S-19500 and as follows. When the diminutive size or lack of suitable surface area prevents routine marking, on the device, of all items required by Specification MIL-S-19500, the following items may be omitted in the following preferred order: color-band type identification (if specified for the device), country of origin, manufacturer's identification. Where only a minimum of items can suitably be marked on the device, first consideration shall be given to marking the complete type designation (see 3.5.1 herein), and then to inclusion of the acceptance date and inspection lot identification. However, all required marking shall be placed on the unit package.

3.5.1 Complete type-designation marking.- Complete type-designation marking of transistors procured on Department of Army contracts, and which have passed Government inspection and comply with all requirements of this specification, shall consist of: "USA-manufacturer's qualification code letters-transistor designation (including any assigned reliability indicator)." The letters "JAN" or any abbreviation therof shall not be used. If any specification waiver has been granted, the combination "USA-manufacturer's qualification code letters" shall not be used to complete the type-designation marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.1.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification and acceptance inspection.- Qualification and acceptance inspection shall be in accordance with requirements in Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. Groups A and B inspection shall consist of the examinations and tests specified in Tables I and II, respectively, herein. Acceptance inspection shall include inspection of Preparation For Delivery (see 5.1 herein).

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4.2.1 Specified LTPD for subgroups. - The LTPD specified for a subgroup in Tables I and II herein shall apply for all of the tests, combined, in the subgroup. (See 6.2 herein.)

4.2.2 Disposition of sample units. - Sample units subjected to Group B, Subgroup 5 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroup 2, 3, 4, 6, and 7 tests may be delivered on the contract or order provided that, after Group B inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.3 Particular examination and test procedures. -

4.3.1 Base Current test. - The specified voltage and current shall be applied to the respective terminals, and the base (significant terminal) current shall then be measured.

4.3.2 Small-Signal Short-Circuit Reverse-Transfer Capacitance test. - Measurement of this characteristic (see 3.2) shall be in accordance with the test set-up shown in Fig. 4 herein. Constant a-c voltage at the specified frequency is applied to the collector, and a measuring resistance is incorporated in the base-to-ground circuit. The a-c voltage developed across this resistance is directly proportional to the capacity being measured: the meter in the test circuit can be calibrated directly in pf so as to permit direct reading of the C_{re} value. The following additional details regarding test conditions, instrumentation, and procedures apply (see Fig. 4 herein):

a. The frequency, 450 kc, is chosen so that the conductive component relative to C_{re} from collector to base is negligible.

b. The 450-kc, a-c, voltage is produced by a crystal oscillator, and is amplified by means of a stabilized RF amplifier. The a-c voltage across the measuring resistance is also amplified by means of a stabilized, 3-stage, measuring amplifier, and is supplied to the meter via a cathode follower and detector.

c. The supply voltages for the transistor under test, and for the various amplifiers, are regulated.

d. For calibration purposes, the transistor to be tested, and the test socket, can be replaced by a calibrating (reference) capacitor.

4.4 Mechanical damage resulting from tests. - Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit as a result of any of the Group A or B tests.

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Table I. Group A Inspection.

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		10				
2071	Visual and mechanical examination	---		---	---	---	---
	Subgroup 2		5				
3036	Collector-to-base cutoff current	Bias Cond. D V_{CB} = -6 Vdc I_E = 0		I_{CBO}	---	-8	uAdc
1/	Base current	V_{CE} = -6 Vdc I_C = 1 mAdc		I_B	-4	-25	uAdc
3076	Forward-current transfer ratio	V_{CE} = -6 Vdc I_C = 1 mAdc		h_{FE}	-40	-250	---
3066	Base-to-emitter voltage	Test Cond. B V_{CE} = -6 Vdc I_B = 1 mAdc		V_{BE}	-210	-330	mVdc
3001	Collector-to-base breakdown voltage	Bias Cond. D I_C = 50 uAdc I_E = 0		BV_{CBO}	-40	---	Vdc
3011	Collector-to-emitter breakdown voltage	Bias Cond. D I_C = 1 mAdc I_B = 0		BV_{CEO}	-20	---	Vdc
3011	Collector-to-emitter breakdown voltage	Bias Cond. B I_C = -50 uAdc R2 = 2 k ohms		BV_{CER}	-40	---	Vdc
3026	Emitter-to-base breakdown voltage	Bias Cond. D I_E = 50 uAdc I_C = 0		BV_{EBO}	-1	---	Vdc
2/	Available power gain	V_{CB} = -6 Vdc I_E = 1 mAdc f = 100 mc Test circuit per Fig. 3 herein		P.G.	10	---	db

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Table I. Group A inspection - (Cont'd).

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 2-(Cont'd)						
2/	Available power gain	V_{CB} = -4 Vdc I_E = 1.5 mAdc f = 100 mc Test current per Fig. 3 herein		P.G.	10	---	db
	Subgroup 3		10				
3261	Unity gain (h_{fe} = 1) frequency	V_{CE} = -6 Vdc I_E = 1 mAdc		f_t	40	---	mc
3206	Small-signal short-circuit forward-current transfer ratio	V_{CE} = -6 Vdc I_C = 1 mAdc f = 1 kc		h_{fe}	60	240	---
3/	Small-signal short-circuit reverse-transfer capacitance	V_{CE} = -6 Vdc I_E = 1 mAdc f = 0.45 mc		C_{re}	---	-2.4	pf
3266	Small-signal short-circuit input conductance	V_{CB} = -6 Vdc I_E = 1 mAdc f = 100 mc Common base; Parallel component		RE_{hib}	---	50	mmhos
3236	Output capacitance	V_{CB} = -6 Vdc I_E = 1 mAdc f = 100 mc		C_{ob}	---	4	pf
3240	Input capacitance	V_{CB} = -6 Vdc I_E = 1 mAdc f = 100 mc		C_{ib}	---	25	pf
4126	Noise figure	Test Cond. A V_{CB} = -6 Vdc I_E = 1 mAdc f = 10.7 mc Source impedance = 150 ohms		NF	---	8	db

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Table I. Group A inspection-(Cont'd).

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3-(Cont'd)						
4126	Noise figure	Test Cond. A V_{CB} = -6 Vdc I_E = 1 mAdc f = 100 mc Source impedance = 68 ohms		NF	---	11	db
	Subgroup 4 4/		20				
5/	High-temperature operation:	T_A = + 60°C, min					
	Collector-to-base cutoff current	Bias Cond. D V_{CB} = -6 Vdc I_E = 0		I_{CBO}	---	50	uAdc
5/	Low-temperature operation:	T_A = 0°C, max					
3076	Static forward-current transfer ratio	V_{CE} = -6 Vdc I_C = 1 mAdc		h_{FE}	15	---	---

1/ See 4.3.1.

2/ Test procedure and computations in accordance with Fig. 3 herein.

3/ See 4.3.2 herein.

4/ For this Subgroup, the sample units subjected to the High Temperature Operation test shall be permitted to return to and be stabilized at room temperature prior to their being subjected to the Low Temperature Operation test.

5/ Measurement shall be made after thermal equilibrium has been reached at the temperature specified.

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Table II. Group B inspection.

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		20				
2066	Physical dimensions			---	---	---	---
	Subgroup 2		15				
2026	Solderability	---		---	---	---	---
1051	Temperature cycling	Test Cond. A except $T_{(high)} = +75° \; {}^{+0°}_{-3°}C$		---	---	---	---
1056	Thermal shock (glass strain)	Test Cond. A		---	---	---	---
1021	Moisture resistance	No initial conditioning		---	---	---	---
	End-point tests:						
3036	Collector-to-base cutoff current	Bias Cond. D $V_{CB} = -6$ Vdc $I_E = 0$		I_{CBO}	---	-12	uAdc
3206	Small-signal short-circuit forward-current transfer ratio	$V_{CE} = -6$ Vdc $I_C = 1$ mAdc f = 1 kc		h_{fe}	30	---	---
3001	Collector-to-base breakdown voltage	Bias Cond. D $I_C = 100$ uAdc $I_E = 0$		BV_{CBO}	-40	---	Vdc
3026	Emitter-to-base breakdown voltage	Bias Cond. D $I_E = 100$ uAdc $I_C = 0$		BV_{EBO}	-1	---	Vdc
	Subgroup 3		15				
2016	Shock	Non-operating G = 500 5 blows of 1 msec ea. in orientations X1, Y1, Z1 (total = 15 blows)		---	---	---	---

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Table II. Group B inspection -(Cont'd).

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3-(Cont'd)						
2006	Constant acceleration (centrifuge)	G = 20,000 Orientations X1, Y1, Z1		---	---	---	---
2046	Vibration fatigue	Non-operating		---	---	---	---
2056	Vibration, variable frequency	---		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 2 above						
	Subgroup 4		15				
1041	Salt atmosphere (corrosion)	t = 96 hrs.		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 2 above						
	Subgroup 5		20				
2036	Lead fatigue	Test Cond. E		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 2 above						
	Subgroup 6		λ=15				
1031	High-temperature life (nonoperating)	T_{stg} = +100°C, min		---	---	---	---

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Table II. Group B inspection-(Cont'd).

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 6-(Cont'd)						
	End-point tests:						
3036	Collector-to-base cutoff current	Bias Cond. D V_{CB} = -6 Vdc I_E = 0		I_{CBO}	---	-16	uAdc
1/	Base current	VCE = -6 Vdc I_C = 1 mAdc		I_B	3	30	uAdc
3001	Collector-to-base breakdown voltage	Bias Cond. D I_C = -100 uAdc I_E = 0		BV_{CBO}	-40	---	Vdc
3026	Emitter-to-base breakdown voltage	Bias Cond. D I_E = -100 uAdc I_C = 0		BV_{EBO}	-1	---	Vdc
	Subgroup 7		λ=15				
1026	Steady state operation life	V_{CB} = -10 Vdc I_E = 10 mAdc		---	---	---	---
	End-point tests:						
	Same as listed under Subgroup 6 above						

1/ See 4.3.1 herein.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following exceptions, are applicable to this specification.

6.2 Re-evaluation or verification inspection.- The LTPD method is exceptionally well suited for inspection at source, since it provides a high degree of assurance (90% confidence) that the lot represented has a proportion defective less than the specified LTPD value. However, the LTPD method is not suitable for inspection performed subsequent to source inspection since it provides, at most, a 10% confidence that the lot represented by a failed sample actually contains a proportion defective in excess of the specified LTPD value. As a result, whenever the quality of a lot is re-evaluated or verified by sampling inspection subsequent to the supplier's satisfactory demonstrations of compliance with the quality requirements, lot disposition should be based on a sampling plan which provides reasonable assurance that any lot rejected contains a proportion defective greater than the specified LTPD or λ value for any individual subgroup. When deemed necessary, the purchase order should specify the detailed criteria for lot disposition.

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, supplement (Army), whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Chief, Standardization Engineering Division, U. S. Army Electronics Materiel Support Agency, Fort Monmouth, New Jersey, attention: SELMS-PSM-3.

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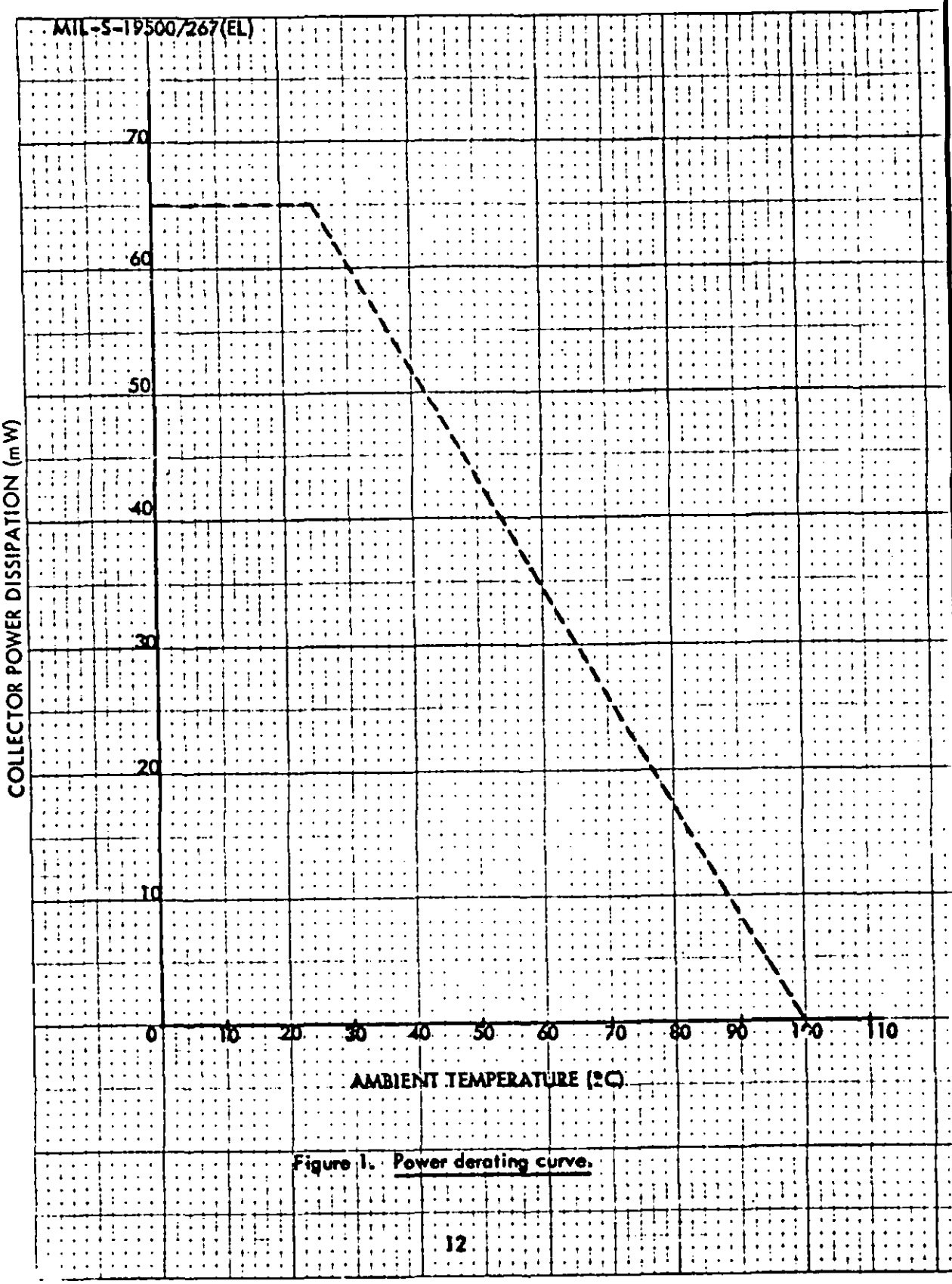
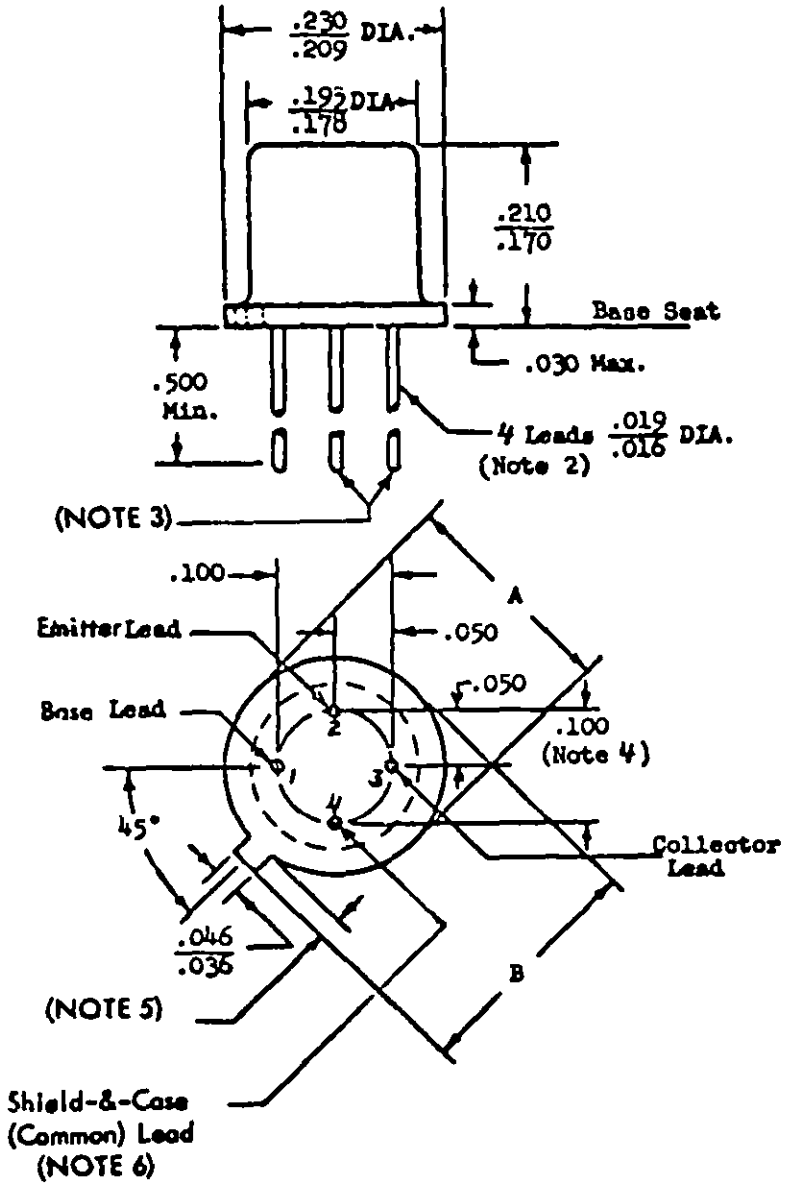


Figure 1. Power derating curve.

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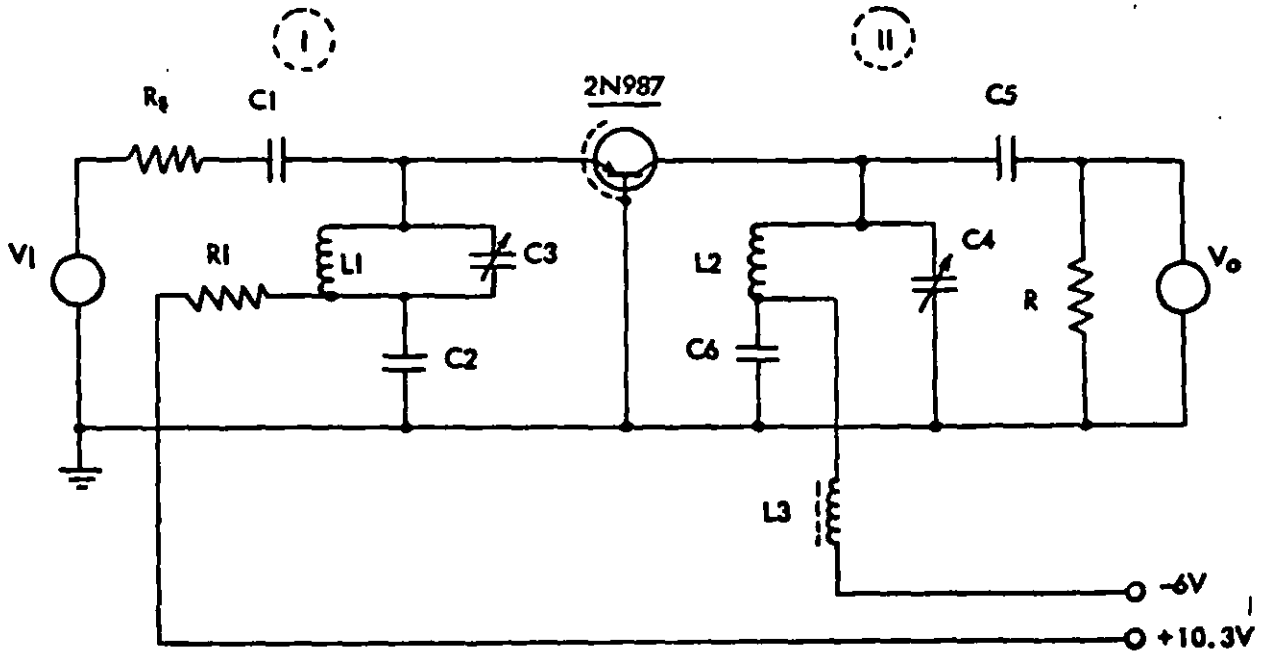


NOTES:

1. All dimensions in inches.
2. The specified lead diameter applies to the zone between .050 and .250 from the base seat. Between .250 and end of lead a maximum of .021 is held. Outside of these zones the lead diameter is not controlled.
3. Recommended: Leads 2 and 3 to be approximately 0.1 longer than leads 1 and 4, to enable easy insertion of unit.
4. Maximum diameter leads at a gaging plane .054 + .001 -.000 below base seat to be within .007 of their true location relative to Max. width tab and to the maximum .230 diameter measured with a suitable gage. When gage is not used, measurement will be made at base seat.
5. Tab length to be .028 minimum -- .048 maximum, and will be determined by subtracting diameter A from dimension B.
6. Lead 4 available for grounding as necessary for proper application of unit in particular circuit.

Figure 2. Outline and dimensions.

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$$\text{Available Power Gain} = \left(\frac{V_o}{V_1}\right)^2 \frac{4\,R_s}{R_L}$$

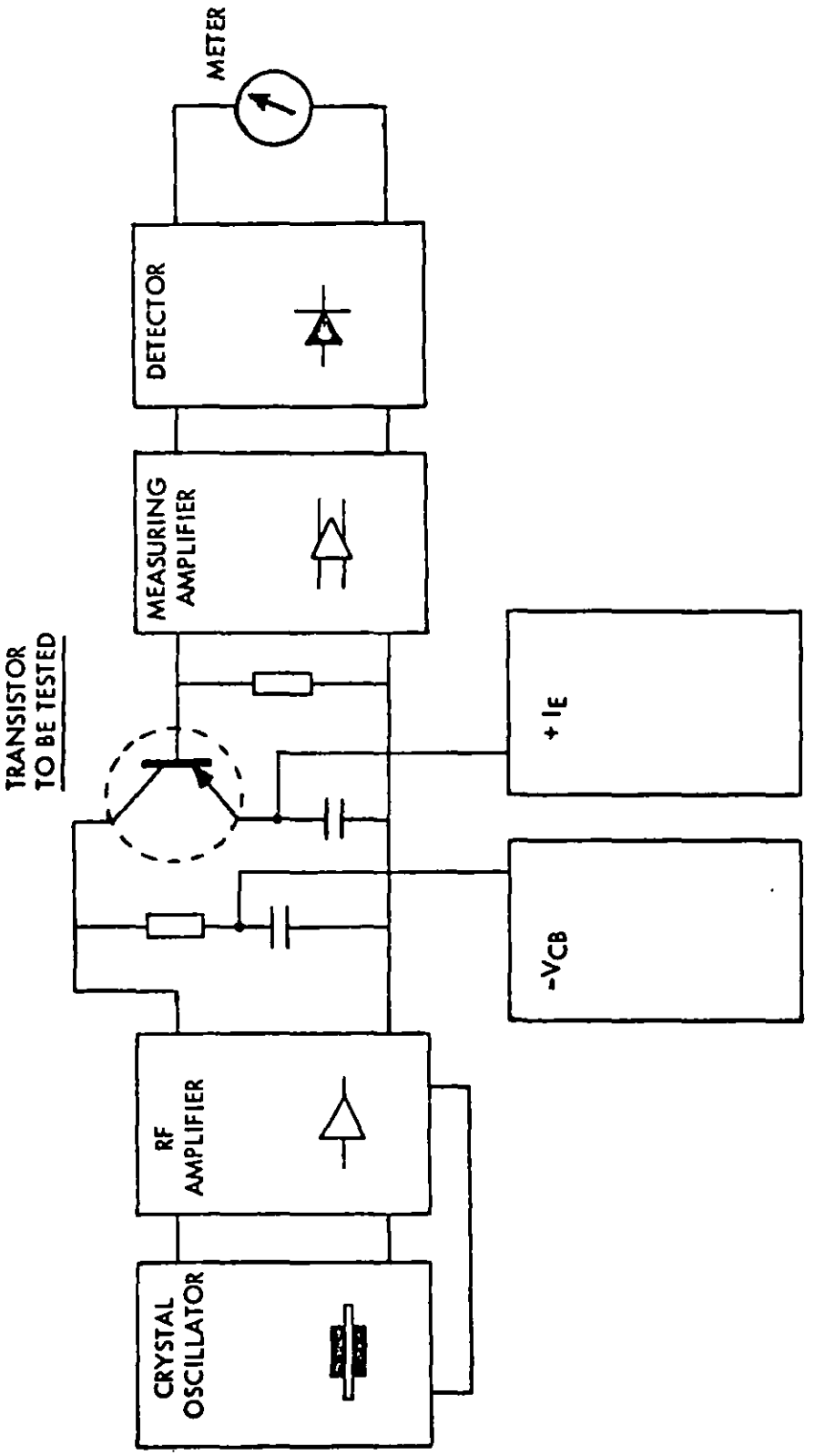
R_1 = 10 kohm
R_s = 60 ohm
R is chosen such that the total impedance R_L of the tuned circuit II equals 3.3 kohm

C_1 = 1500 uuf
C_2 = 820 uuf
C_3 = 5-25 uuf var.
C_4 = 2-10 uuf var.
C_5 = 220 uuf
C_6 = 820 uuf

L_1 = 0.18 uH
L_2 = 0.18 uH
L_3 = ferrite bead

Figure 3. Available Power Gain test circuit.

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NOTE: Refer to circuit details in 4.3.2 herein.

Figure 4. Schematic of test circuit for measurement of "C_{re}".

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